Business Analytics Case Study

Targeted Marketing Strategy for Bank Term Deposit Subscriptions

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Dataset: Bank Marketing (UCI / Kaggle)

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1. Introduction

This report presents a business analytics case study built on a bank telemarketing dataset. The objective of the case study is to move beyond simple data exploration and answer a specific business question with a defensible, quantified recommendation. The full analysis was conducted in PostgreSQL, following a structured process of data auditing, cleaning, business-driven querying, and a cost-savings model built to translate the findings into a dollar-based business case.

Every conclusion in this report is supported by SQL query output, and every assumption used in the cost calculations is stated explicitly rather than hidden inside the numbers.

2. Business Problem Statement

A bank runs outbound telemarketing campaigns to sell term deposits. Each contact attempt carries a real cost in call center time, yet only a fraction of customers contacted actually subscribe. Calling every customer in the same way, regardless of how likely they are to convert, wastes budget on low-probability contacts and reduces the efficiency of the entire campaign.

The business question this analysis answers is: which customer segments and campaign conditions produce the highest conversion probability, and how much could the bank save by prioritizing those segments instead of contacting every customer with equal effort?

This reframes the task from a general exploration of customer behavior into a targeting and cost-efficiency problem, which is the lens the rest of this report uses.

3. Dataset Description

The dataset contains 11,162 customer contact records from a bank telemarketing campaign, with 17 original fields covering customer demographics (age, job, marital status, education), financial standing (account balance, housing loan, personal loan), and campaign details (contact method, month, number of contacts, days since last campaign, previous campaign outcome, and the subscription result).

The audit confirmed the dataset has no missing values and no duplicate rows. However, two structural characteristics required correction before the analysis could be trusted.

1. The target variable is artificially balanced. The dataset shows a 47.4 percent subscription rate, while real-world bank telemarketing campaigns typically convert around 11 percent of contacts. This version of the dataset has been resampled for modeling convenience, so all cost projections in this report are reweighted against the realistic 11 percent baseline rather than the dataset's own inflated rate.

2. The call duration field reflects information only available after a call ends, so it cannot inform who to call before dialing. It was excluded from all targeting and driver analysis and is referenced only as a separate observation on call quality.

4. Data Analysis Process

The analysis followed four stages, each completed and verified before moving to the next.

4.1 Audit

Row count, null values, duplicate records, target balance, placeholder or unknown values, and a check on the duration leakage variable were all verified first, before any cleaning was performed.

4.2 Cleaning

A prior_contact_flag field was created to correctly represent the 74.6 percent of customers who had never been contacted in a previous campaign, replacing three separate unknown or placeholder values (pdays equal to negative one, previous equal to zero, and poutcome equal to unknown) with a single accurate category rather than treating them as missing data. Job titles and month values were also standardized to full, consistent lowercase labels.

4.3 Business-Driven Querying

Twelve SQL queries were written to answer specific business questions, covering conversion rate by job, education, marital status, housing loan status, personal loan status, contact method, month, prior contact history, campaign contact count, age band, and a combined job and prior-contact ranking used to build the final targeting list.

4.4 Cost-Savings Modeling

Two targeting approaches were tested and compared for reliability before selecting one as the basis for the business recommendation, followed by a capped-contact model that translates the findings into a specific, actionable cost-saving change.

5. Key Findings

Variable	Finding
Job type	Student (74.7%) and retired (66.3%) convert highest; blue-collar (36.4%) converts lowest despite being the second-largest segment.
Education	Conversion rises steadily from primary (39.4%) to secondary (44.7%) to tertiary (54.1%).
Marital status	Married customers are the largest segment (6,351 contacts) but convert lowest (43.4%); single customers convert highest (54.3%).
Housing loan	Non-holders convert at 57.0% versus 36.6% for holders, a 20-point gap.
Personal loan	Non-holders convert at 49.5% versus 33.2% for holders.

Contact method	Cellular (54.3%) and telephone (50.4%) convert well; unknown method converts at only 22.6%, likely a data logging gap.
Prior contact	Previously contacted customers convert at 67.1% versus 40.7% for those never contacted before, the strongest single predictor found.
Campaign contact count	Conversion falls steadily from 53.4% at one contact to 26.8% at seven or more contacts.
Age	60+ converts highest (76.9%); the working-age middle (30-49) converts lowest (40-44%).

Across every variable tested, the same pattern repeats: customers with fewer competing financial obligations and prior familiarity with the campaign convert at meaningfully higher rates than the campaign's highest-volume segments, which tend to be the lowest converting.

6. Business Insights

The individual findings above combine into one consistent business insight. The segments the campaign currently calls the most, married customers, blue-collar workers, and customers with no prior contact history, are also the segments that convert the least. This means a large share of the campaign's call volume and cost is currently concentrated in its least productive segments, rather than being weighted toward the segments most likely to say yes.

Testing a more detailed, multi-variable targeting list (job, prior contact, loan status, and contact count combined) produced a marginally higher conversion rate at its top end, but on customer groups as small as thirty to seventy people. A rate built on a sample that small is not reliable enough to act on with confidence, so the simpler two-variable targeting list (job type combined with prior contact history) was chosen as the basis for the business recommendation. This decision was made deliberately, favoring a result that is strong and statistically trustworthy over one that is marginally higher but too small to defend.

7. Recommendations

1. Prioritize the targeted segments earlier in the calling cycle. Customers matching the qualifying job type and prior contact combinations convert at an estimated 30.78 dollars per conversion, compared to 53.20 dollars per conversion for the rest of the customer base, a 42 percent efficiency gain. These segments should be called first when call center capacity is limited.

2. Cap contact attempts at three calls per customer for the lower-converting segment. Conversion drops sharply after the third call, from 53.4 percent at one contact to 26.8 percent at seven or more. Capping contact attempts is estimated to save approximately 32,775 dollars by avoiding 6,555 low-yield repeat calls. Approximately 16 percent of this segment's conversions occurred on a fourth call or later, so this cap should be piloted on a subset of the campaign before full rollout to confirm the actual conversion impact.

3. Reassess how the offer is positioned for customers carrying housing or personal loans. These customers convert at roughly half the rate of customers without existing loan obligations. Rather than simply deprioritizing them, the bank could test an adjusted offer, such as a lower minimum deposit or more flexible terms, that better fits their existing financial commitments.

4. Audit the contact-logging process for the unknown contact method category. This group converts at less than half the rate of cellular or telephone contacts, which more likely reflects a gap in how contact attempts are recorded than a genuine behavioral pattern, and should be corrected before this variable is used in future targeting decisions.

5. Validate all findings against the bank's own unweighted campaign data before committing budget. The dataset used in this analysis is artificially balanced at a 47.4 percent conversion rate, and every dollar figure in this report has been reweighted against an assumed realistic 11 percent baseline. This produces a directionally sound business case, but the bank should confirm segment-level lift against its actual, non-resampled campaign records before finalizing budget decisions.

8. Conclusion

This analysis shows that a small set of customer characteristics, occupation, prior contact history, and existing loan obligations, consistently explain who is more likely to subscribe to a term deposit. The campaign's current calling pattern does not reflect this: its highest-volume segments are also its least efficient. Shifting call priority toward the identified high-converting segments, while capping wasted repeat contact within the lower-converting segment, offers the bank a realistic and measurable path to a more cost-efficient campaign, without abandoning any part of its existing customer base.

Every recommendation in this report is tied directly to a specific, verifiable finding in the underlying SQL analysis, and every assumption used to translate those findings into dollar figures has been stated openly rather than built silently into the numbers.